

DATA PREPROCESSING IN DATA SCIENCE

AIM

To understand the importance of data preprocessing in data science and to prepare a raw data set for machine learning.

PROGRAM

```
import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split

df = pd.read_csv("pre-process_datasample.csv")
print(df)
df.info()

print("Mode of Country :", df.Country.mode()[0])
df['Country'] = df['Country'].fillna(df.Country.mode()[0])
df['Age'] = df['Age'].fillna(df.Age.median())
df['Salary'] = df['Salary'].fillna(round(df.Salary.mean()))
print(df)
df.info()

print(pd.get_dummies(df.Country))
dummies = pd.get_dummies(df.Country)
updated_dataset = pd.concat([dummies, df.iloc[:, [1, 2, 3]]], axis=1)
print(updated_dataset)

updated_dataset['Purchased'] = updated_dataset['Purchased'].replace(
    {'No': 0, 'Yes': 1})
print(updated_dataset)

# Splitting into training and testing sets
X = updated_dataset.iloc[:, :-1]
y = updated_dataset.iloc[:, -1]
x_train, x_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=0)
print("Training set :", x_train.shape, " Testing set :", x_test.shape)
```

OUTPUT

```
   Country  Age  Salary  Purchased
0  France  44.0  72000.0         No
1   Spain  27.0  48000.0         Yes
2  Germany  30.0  54000.0         No
3   Spain  38.0  61000.0         No
4  Germany  40.0     NaN         Yes
5  France  35.0  58000.0         Yes
6   Spain   NaN  52000.0         No
7  France  48.0  79000.0         Yes
8     NaN  50.0  83000.0         No
9  France  37.0  67000.0         Yes
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Country      9 non-null        str
1   Age          9 non-null        float64
2   Salary       9 non-null        float64
3   Purchased    10 non-null       str
dtypes: float64(2), str(2)
memory usage: 452.0 bytes
```

```

Mode of Country : France
  Country  Age  Salary Purchased
0  France  44.0  72000.0        No
1   Spain  27.0  48000.0        Yes
2  Germany 30.0  54000.0        No
3   Spain  38.0  61000.0        No
4  Germany 40.0  63778.0        Yes
5   France 35.0  58000.0        Yes
6   Spain  38.0  52000.0        No
7   France 48.0  79000.0        Yes
8   France 50.0  83000.0        No
9   France 37.0  67000.0        Yes
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Country     10 non-null     str
1   Age         10 non-null     float64
2   Salary      10 non-null     float64
3   Purchased   10 non-null     str
dtypes: float64(2), str(2)
memory usage: 452.0 bytes
  France  Germany  Spain
0   True    False  False
1  False    False   True
2  False     True  False
3  False    False   True
4  False     True  False
5   True    False  False
6  False    False   True
7   True    False  False
8   True    False  False
9   True    False  False
  France  Germany  Spain  Age  Salary  Purchased
0   True    False  False  44.0  72000.0        No
1  False    False   True  27.0  48000.0        Yes
2  False     True  False  30.0  54000.0        No
3  False    False   True  38.0  61000.0        No
4  False     True  False  40.0  63778.0        Yes
5   True    False  False  35.0  58000.0        Yes
6  False    False   True  38.0  52000.0        No
7   True    False  False  48.0  79000.0        Yes
8   True    False  False  50.0  83000.0        No
9   True    False  False  37.0  67000.0        Yes
  France  Germany  Spain  Age  Salary  Purchased
0   True    False  False  44.0  72000.0         0
1  False    False   True  27.0  48000.0         1
2  False     True  False  30.0  54000.0         0
3  False    False   True  38.0  61000.0         0
4  False     True  False  40.0  63778.0         1
5   True    False  False  35.0  58000.0         1
6  False    False   True  38.0  52000.0         0
7   True    False  False  48.0  79000.0         1
8   True    False  False  50.0  83000.0         0
9   True    False  False  37.0  67000.0         1
Training set : (8, 5)  Testing set : (2, 5)

```

RESULT

Thus the data set was preprocessed: missing values were filled, the categorical column was encoded, the label was converted to numbers and the data was split into training and testing sets.